Do not consider possible user error cases in this series

Solution of Exercise 2.07:

```
Algo Exo2_7
Var
  A, B, C, D, Max : Integer
Begin
    Read (A, B, C, D)
    Max ← 0
    If A mod 6 = 0 Then
      Max ← A
    EndIf
    If B > Max and B mod 6 = 0 Then
      Max ← B
    EndIf
    If C > Max and C mod 6 = 0 Then
      Max ← C
    EndIf
    If D > Max and D mod 6 = 0 Then
      Max ← D
    EndIf
    If Max = 0 Then
      Write("no number entered is a multiple of 6")
    Else
      Write(Max)
    EndIf
End
```

Solution of Exercise 2.08_d:

```
Algo Exo2_8_d
Var
  a, b, c, Sol1, Sol2, Delta : Real
Begin
  Read(a,b,c)
  If a = 0 Then
    If b = 0 Then
      If c = 0 Then
        Write("Infinite number of solutions")
      Else
        Write("No solution")
      EndIf
    Else
      Sol1 ← -c/b
      Write("One solution :", Sol1)
    EndIf
  Else
    Delta ← b*b – 4*a*c
    If Delta < 0 Then
      Write("No solution in R")
    Else
      If Delta = 0 Then
        Sol1 ← -b/(2*a)
        Write("Dual solution :", Sol1)
      Else
        Sol1 ← (-b+sqrt(delta))/(2*a)
        Sol2 ← (-b-sqrt(delta))/(2*a)
        Write("Two solutions :", Sol1, " " , Sol2)
      EndIf
    EndIf
  EndIf
End
```

For a=0, b=2 and c=5, **One Sulution : -2.5**

For a=0, b=0 and c=6, **No solution**

For a=0,b=0 and c=0, **Infinite number of solutions**

Solution of Exercise 2.09:

```
Algo Ex2_9_a
Var
NCI, NCD, NE, NN, PI, PD, PE, PPD, PaP : Integer
Begin
        Read(NCI, NCD, NE, NN)
        If NCI < 3 Then
                PI ← NCI * 7500
        Else
                PI ← NCI * 6000
        EndIf
        If NCD<4 Then
                PD← NCD * 9500
        Else
                PD← NCD * 8000
        EndIf
        If NE=0 Then
                PE← 0
        Else
                PE←(NE – 1) * 3000
        EndIf
        PPD ←(NE + NCI + NCD *2) * 300
        PaP← (PI + PD + PE + PPD) * NN
        Write ("The price to pay is ", PaP)
End
```

```
Algo Ex2_9_b
Var
NCI, NCD, NE, NN, PI, PD, PE, PPD, PaP : Integer
Begin
        Read(NCI, NCD, NE, NN)
        If NCI < 3 Then
                PaP ← NCI * 7500
        Else
                PaP ← NCI * 6000
        EndIf
        If NCD<4 Then
                PaP ← PaP + NCD * 9500
        Else
                PaP ← PaP + NCD * 8000
        EndIf
        If NE ≠ 0 Then
                PaP ← PaP + (NE – 1) * 3000
        EndIf
        PaP ← PaP + (NE + NCI + NCD *2) * 300
        Write ("The price to pay is ", PaP*NN)
End
```

Solution of Exercise 2.10:

```
Algo Exo2_10
Var
  M, A : Integer
Begin
    Read (M, A)
    Case M of
      1, 3, 5, 7, 8, 10, 12 : Write ("31 days")
      4, 6, 9, 11 : Write ("30 days")
      2 : If (A Mod 4 = 0 and A Mod 100 ≠ 0) or A Mod 400 = 0 Then
            Write ("29 days")
          Else
            Write ("28 days")
          EndIf
    EndCase
End
```

Solution of Exercise 2.11:

```
Algo Ex2_11_a1
Var
  h, m : Integer
Begin
  Read(h,m)
  If m>=0 and m<59 and h>=0 and h<=23 Then
    Write ("In a minute it will be",h,":",m+1)
  EndIf
  If m=59 and h>=0 and h<23 Then
    Write ("In a minute it will be",h+1,":00")
  EndIf
  If m=59 and h=23 Then
    Write ("In a minute it will be 00:00")
  EndIf
End
```

```
Algo Ex2_11_a2
Var
  h, m : Integer
Begin
    Read(h,m)
    m ← m + 1
    If m=60 Then
      m←0
      h← h +1
      If h=24 Then
        h← 0
      EndIf
    EndIf
    Write ("In a minute it will be",h,":",m)
End
```

```
Algo Ex2_11_b
Var
  h, m : Integer
Begin
    Read(h,m)
    m ← m + 8
    If m ≥ 60 Then
      m← m - 60
      h← h +1
      If h=24 Then
        h← 0
      EndIf
    EndIf
    Write ("In 8 minutes it will be ",h,":",m)
End
```

Solution of Exercise 2.04:

Algo Exo2_4_3	Algo Exo2_4_1

Algo Exo2_4_3
Var
P, T, BMI: Real
Begin
Write ("Please enter your weight in kg, then your height in meters ")
Read (P, T)
BMI ← P / (T*T)
Write(BMI)
If BMI < 18.5 **Then**
Write ("Thinness")
EndIf
If BMI ≥ 18.5 and BMI ≤ 25 **Then**
Write ("Normal range")
EndIf
If BMI > 25 **Then**
Write ("Overweight")
EndIf
End
	Algo Exo2_4_1
	Var
	P, T, BMI : Real
	Begin
	Write ("Please enter your weight in kg, then your height in meters ")
	Read (P, T)
	BMI ← P / (T*T)
	Write(BMI)
	If BMI < 18.5 **Then**
	Write ("Thinness")
	Else
	If BMI ≤ 25 **Then**
	Write ("Normal range")
	Else
	Write ("Overweight")
	EndIf
	EndIf
	End

For P=60 and T=1.60, BMI= $\boxed{23.4375}$

Normal range

For P=75 and T=1.65, BMI= $\boxed{27.5482}$

Overweight

For P=55 and T=1.75, BMI= $\boxed{17.9592}$

Thinness

Exercise 2.13:

☐	☐	☐	☐	☐

Algo Ex_2_13_a
Var
A, B, C : Real
Begin
Read (A, B)
C ← A / B*B
If C < 10 **Then**
Write (A)
EndIf
If C≥10 and C≤20 **Then**
Write (C)
EndIf
If C > 20 **Then**
Write (B)
EndIf
End
	Algo Ex_2_13_b
	Var
	A, B, C : Real
	Begin
	Read (A, B)
	C ← A / (B*B)
	If C ≤ 10 **Then**
	Write (A)
	EndIf
	If C>10 and C≤20 **Then**
	Write (C)
	EndIf
	If C > 20 **Then**
	Write (B)
	EndIf
	End
		Algo Ex_2_13_c
		Var
		A, B, C : Real
		Begin
		Read (A, B)
		C ← A / B*B
		If C < 10 **Then**
		Write (A)
		Else
		If C ≤ 20 **Then**
		Write (C)
		Else
		Write (B)
		EndIf
		EndIf
		End
			Algo Ex_2_13_d
			Var
			A, B, C : Real
			Begin
			Read (A, B)
			C ← A / B*B
			If C > 20 **Then**
			Write (B)
			Else
			If C < 10 **Then**
			Write (A)
			Else
			Write (C)
			EndIf
			EndIf
			End
				Algo Ex_2_13_e
				Var
				A, B, C : Real
				Begin
				Read (A, B)
				C ← A / B*B
				If C ≤ 20 **Then**
				Write (C)
				Else
				If C < 10 **Then**
				Write (A)
				Else
				Write (B)
				EndIf
				EndIf
				End